2023 Ground Truth Document Classification CABR Version 1 Data-Envelope Version 1

- DataEnvelope
  - status
    - under construction
  - Level 1: Basic Information
    - 1.1. Title of Data-Envelope
      - 2023 Ground Truth Document Classification CABR Version 1 Data-Envelope Version 1
    - 1.2. Contact Details
      - Name
        - Data & Collections department, Huygens Institute
      - roleInProject
        - Data Curation
      - Email
        - data@huygens.knaw.nl
    - 1.3. Data Envelope Dates
      - Data-Envelope Creation Date
        - 2025-03-25
      - Data-Envelope Completion Date
        - 2025-07-03
    - 1.4 Author of Data-Envelope
      - Name
        - Mari Wigham
      - orcidID
        - 0000-0002-2784-1921
      - email
        - data@huygens.knaw.nl
    - 1.4 Author of Data-Envelope
      - Name
        - Lotte Belice Baltussen
      - orcidID
        - 0009-0003-3961-4325
      - email
        - data@huygens.knaw.nl
  - Level 2: Basic Metadata
    - 2.1 Snapshot
      - Dataset Title
        - 2023 Ground Truth Document Classification
      - Version
        - 1
      - Description
        - This dataset contains ground truth (GT) that was created in order to train a document classification model for the Central Archives of Special Jurisdiction (CABR, 2.09.09),

as part of the project War in Court (Oorlog voor de Rechter) [1]. ABOUT THE ARCHIVE This archive is held within the National Archives of the Netherlands. It contains files of around 425,000 people that were investigated within the Special Jurisdiction. Among them were people who had been members of the National Socialist Movement (NSB), or who were suspected of crimes such as betrayal or fighting in the German army. But also people who were investigated and found to be innocent. The archive contains around 30 million pages. Digitisation is ongoing and will finish in 2027. The archive contains a lot of different types of documents. They are also not grouped together within the files, and users are mostly interested in specific documents, such as verdicts and police reports. Furthermore, they have trouble reconstructing the judicial timeline, which document classification can facilitate. We therefore created this GT set to train a document classification model. ABOUT THIS GROUND TRUTH SET This is the first GT that was created. We classified on the level of one scan. We were able to look at scans before and after the one that we were classifying, in order to provide more context and thus a more accurate classification. In total, we classified 7,472 pages. The GT gives an insight in what type of documents the files in this archive consists of, although it is skewed towards cases that were treated in the Supreme Court of the Special Jurisdiction. The underlying documents are not part of the GT due to legal restrictions. We started creating the ground truth mid-October 2023, and finished the process late November 2023. We will create and share more ground truth as a separate data set. [1] You can find more information about the project at https://oorlogvoorderechter.nl/about-the-project/.

Genre
: Document

Genre
: Legal cases and case notes

Topic Classification
: Law

Topic Classification
: Modern and contemporary history

Topic Classification
: Criminal law and Criminology

Other Topic
: World War Two

Languages
: English

Geographical Coverage
: Netherlands

Other Geographical Coverage
: Germany

Other Geographical Coverage
Belgium
Other Geographical Coverage
Denmark
Other Geographical Coverage
France
TemporalCoverage
Temporal Coverage Start Date
1934
Temporal Coverage End Date
1952
Additional Notes
Scans of Dutch documents from the 1940’s and 1950’s make up the bulk of the ground truth. These documents are mostly evidence from before and during WW2, and legal documents from directly after WW2. Therefore, the ground truth would not be ideal to train a set of documents from outside this period and context.

2.3 Creators and Contributors

Publishing Organisation
Name
Huygens Institute for Dutch History and Culture and its predecessors
ROR ID
https://ror.org/04x6kq749
Type
Academic/Research Insitution
Creators
Name
Lotte Belice Baltussen
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Organisation
Huygens Institute for History and Culture of the Netherlands
ROR ID
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Role
Data Curation
Funding Source
Name
Ministerie van Onderwijs Cultuur en Wetenschap
ROR ID
https://ror.org/02x3w5g21
Funding Source
Name
Ministerie van Volksgezondheid, Welzijn en Sport
ROR ID
https://ror.org/041evnj42
Funding Source
Name

- Ministerie Justitie en Veiligheid
- ROR ID
  - https://ror.org/04nzhmg91
- 2.4 Distribution
  - Download
    - File Size
      - 1
- 2.5 Licenses and Accesses
  - Lincensing Information
    - Identifier
      - CC0 1.0 Universal
    - URL
      - https://creativecommons.org/publicdomain/zero/1.0/deed.en
  - Access
    - Access Level
      - Open Access
    - Access Restricted
      - Contact Email
        - data@huygens.knaw.nl
- 2.6 Dataset Version and Maintenance
  - Version
    - Current Version
      - 1.0
  - Maintenance
    - Maintenance Status
      - Limited Maintenance
  - Maintenance Plan
    - Updates
      - The Huygens Institute, and the War in Court team specifically, will carry out updates up until the end of 2027, the final year of the project. New ground truth will be created in order to improve the ML model for document classification. These will be published as separate sets, but can be combined. This specific set itself will not be expanded, but metadata about it can be updated.
    - Feedback
      - data@huygens.knaw.nl

Level 3: Data

- 3.1 Data Resource Description
  - Name of Resource
    - 2023 Document classification GT anonymised - Document classification GT and 2023 Document classification GT anonymised - GT addendum key documents
  - Description
    - This spreadsheet lists ID's of the scans, with the categories and subcategories of document types which were assigned to them as ground truth by hand. The scan numbers themselves have been replaced by ID's, because the scan numbers could be used to find out

which person or persons the classified scans relate to. BASIS OF THE GROUND TRUTH The set of scans used for the ground truth (GT) is not representative for the entire CABR. At the time of the GT creation, the vast majority of scans were of the ‘Bijzondere Raad van Cassatie’, comparable to the Supreme Court. There are collections of many other organisations in the CABR, which also contain different documents not available (enough) in the set we worked with. The set consists of a random sample of two datasets: 4,755 scans taken from a set of the ‘Bijzondere Raad van Cassatie’. 2,100 scans come from the TRIADO [1] project (the War in Court predecessor). The TRIADO set contains a sample from the entire CABR. At the time of sampling, there were 181,580 scans in the ‘Bijzondere Raad van Cassatie’ set, and 155,775 TRIADO scans. The random sample was related to the scanning timeframe of the CABR. The TRIADO dataset was already created and the project itself was completed. The ‘Bijzondere Raad van Cassatie’ scans only represent a subset of the total part of this part of the archive. There are files of many other organisations in the CABR. In other words: the random sample was created based on the scans available at the time. However, the CABR contains an estimated 30 million pages, so in order to have a fully representative sample of the documents in the archive, we have created more ground truth after finishing this first set. Scanning is ongoing and will not be completed until 2027. CREATING THE GROUND TRUTH Each scan in the dataset was manually annotated by the ground truth creators. We annotated individual scans of documents in a file. This file is related to a person. Each row in the dataset represents information related to one scan. The scans themselves are not part of the dataset, due to legal reasons. Each annotated scan (instance) is a row in a CSV file. The metadata fields are described the next section. There are some relations between instances. They can belong to the same file, or to the same person in cases where there are multiple files about that person. This is not made explicit in the data. The annotators had access to the scans in our closed web environment, in order for everyone working on the ground truth creation had access to the scans. The scans are not publicly available due to legal restrictions. Each scan was assigned to at least one top-level category (max. 3) and up a subcategory. The categories used are listed in the tab 'Document types - overview', and the hierarchical structure in '2023 Document classification GT anonymised - Hierarchical structure document'. The GT follows this structure. An example of a top-level category is Printed document (Gedrukt document), and an example of a subcategory is Verdict (Uitspraak). In some

cases, it was not possible to determine a document subcategory. In those cases, all of the columns ‘Subcategorie 1’, ‘Subcategorie 2’, ‘ Subcategorie 3’ are empty. We supplemented the random GT with key documents that we wanted the document classification model to focus on. This is documented in 'GT addendum key documents'. It follows the same structure as random GT in the tab 'GT document classification'. [1] You can find more information about TRIADO at https://www.oorlogsbronnen.nl/nieuws/veelbelovende-resultaten-onderzoek-naar-digitaal-doorzoekbaar-maken-cabr.

Format
csv
Size
1
Date
2023-10-02
Language(s)
Dutch
Data Subject(s)
Subject(s)
Data about places and object
Data Modality
Modality
Tabular Data

3.2 Data Fields

dataField
Data Field Name
ID
Data Field Type
integer
Description of the Data Field
The ID of the scan. The original scan name has been replaced with an ID, to prevent the possibility to match the type of document classified to a specific person in the archive.

dataField
Data Field Name
Subcategorie 1
Description of the Data Field
This is the classification - if available - for this document in the first sublevel of categories. In some cases, it was not possible to determine a document subcategory. In those cases, the columns ‘Subcategorie 1’, ‘Subcategorie 2’, ‘ Subcategorie 3’ are all empty.

dataField
Data Field Name
Subcategorie 2
Description of the Data Field

This is the classification - if available - for this document in the second sublevel of categories. In some cases, it was not possible to determine a document subcategory. In those cases, the columns ‘Subcategorie 1’, ‘Subcategorie 2’, ‘ Subcategorie 3’ are all empty.

dataField

Data Field Name

Subcategorie 3

Description of the Data Field

This is the classification - if available - for this document in the third sublevel of categories. In some cases, it was not possible to determine a document subcategory. In those cases, the columns ‘Subcategorie 1’, ‘Subcategorie 2’, ‘ Subcategorie 3’ are all empty.

dataField

Data Field Name

Subcategorie 4

Data Field Type

string

Description of the Data Field

This is the classification - if available - for this document in the fourth and final sublevel of categories. In some cases, it was not possible to determine a document subcategory. In those cases, the columns ‘Subcategorie 1’, ‘Subcategorie 2’, ‘ Subcategorie 3’ are all empty.

dataField

Data Field Name

Afkomstig uit

Data Field Type

string

Description of the Data Field

Indicates whether the classified scan was digitised as part of the predecessor project TRIADO or of the Oorlog voor de Rechter (OVDR) project. The TRIADO set contains a sample from the entire CABR, whereas the OVDR scans at the time of this ground truth creation only contained scans from the Bijzondere Raad van Cassatie. This is comparable to the Supreme Court. There are many other types of organisations in the CABR, that have filed in which the types and number of documents differ from that of the Bijzondere Raad van Cassatie. So, these scans are not representative of the entire archive.

dataField

Data Field Name

Top-level - altijd toepasbaar 1, 2, 3

Data Field Type

string

Description of the Data Field

This is the high-level classification of a document. It can only consist of these six values: Formulier (Form) Fotomateriaal (Photographic material) Gedrukt document (Printed document) Getypt document (Typed document) Handgeschreven document (Handwritten document) Lege pagina (Empty page) In some cases, it is possible to assign multiple top-level categories. In those cases, the instruction was to fill out the most prominent category first, and adding the other ones in ‘Top-level - altijd toepasbaar 2’, ‘ Top-level - altijd toepasbaar 3’. 'Top-level - altijd toepasbaar 1' should always be filled out.

dataField

Data Field Name

GT creator

Data Field Type

string

Description of the Data Field

The anonymised indication of which person from which organisation classified the scan. People from three project partners annotated scans: Huygens Institute National Archives (NA) WO2Net

3.3 Data Point Examples

Typical Example

Description

A typical data point is when Top-level - altijd toepasbaar 1 and Top-level - altijd toepasbaar 2 are both filled out, which Getypt document and Gedrukt document respectively, and where Subcategorie 3 is Proces-verbaal - verklaringen. The reason is that within the Special Jurisdiction, the basis of many of the research done on suspects of collaboration entailed holding interviews with both suspects and witnesses. These interviews were written down and very often these minutes were then typed out on partly pre-printed documents. See for instance ID 452.

Atypical Example

Description

An atypical data point concerns empty pages (Lege pagina). In those cases, it's often impossible to add a sub-level category. See for instance ID 37.

3.4 Errors, Noise, and Redundancies

Description

In some cases, we were unsure which document category to assign. In those cases, the subcategories are: Nog uitzoeken - bewijsmateriaal (To be determined - evidence) Nog uitzoeken - documentatie rechtsgang (To be determined - documentation legal process) Nog uitzoeken - persoonlijk materiaal (To be determined - personal document)

- 3.5 External Resources
  - Resources
    - Yes
  - External Resources Description and Links
    - There are links to our closed environment in which we could view the scans. These are not accessible.
- 3.6 Annotation and Labeling
  - Annotation Characteristics
    - annotationWorkforceType
      - Human Annotations (Expert, Employees)
    - description
      - All annotations have been made by humans.
    - totalAnnotations
      - 7472
- 3.7 Social Impact, Sensitivity, and Biases
  - Safety Consideration
    - Does the resource contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?
      - Yes
    - If yes or you are not sure, please specify:
      - The archive contains evidence related to the Nazi occupation in the Netherlands and the roles Dutch individuals played in it. The dataset contains photos from World War II that depict swastikas and people giving the nazi salute. There are terms and slurs that will no longer be considered acceptable. There is also a lot of information about victims of the Nazi persecution. Furthermore, there are reports by coroners, pictures of dead bodies and descriptions of executions in the dataset. This data is however not present in this Ground Truth dataset.
  - Confidentiality
    - confidentialityBin
      - Yes
    - Specify Measures
      - We masked the scan filesnames, because they contain both the inventory number and the scan number. This can be traced to an individual.
  - biases
    - Known Biases in the Resource
      - The archive itself contains files of around 425,000 people suspected of collaboration after World War II. Around 13% appeared in court, and another 19% did not have to appear in court but received sanctions from a special public prosecutor. In about 329,000 cases, the person was not punished. For example, if a person’s whereabouts were unknown, if someone had passed away, if the offenses were too minor for a punishment, or if the person was innocent. The reasons why people were suspected in the first place are not always known, and could

be rooted in personal feuds. The scans that were annotated for this ground truth were mostly sampled from files of people that appeared in front of a court. This does not represent the majority of the archive, since only 13% of the suspects appeared in front of a court.

Steps taken to Mitigate/Reduce Biases

We didn't take steps to mitigate biases in this round of ground truth. We will however in the future work with more representative samples of the archive, once more has been digitised.

Sensitive Attributes

Sensitive Human Attributes

Others (please specify)

Specify

The CABR by default contains personal information, since the archive consists of files about people suspected of collaboration with the Nazis during World War II. This information is not present in this dataset, but in the archive this dataset is based on.

Ethical Review

Review

No

3.9 Digitisation

Digitalisation Pipeline

See Data Resource Description.

Level 3: Data

3.1 Data Resource Description

Name of Resource

2023 Document classification GT anonymised - Document types overview

Description

This lists the document types used in the classification of the documents. There is a link for document types that are described in the WW2-Thesaurus. This is a validated, thesaurus of key terms for the thematic retrieval of (digital) sources from and about the Second World War. The thesaurus contains terms relating to events, places, concepts, and objects. See also: https://www.niod.nl/en/collections/ww2-thesaurus.

Format

csv

Size

1

Date

2023-10-02

Language(s)

Dutch

Data Subject(s)

Subject(s)

Data about places and object

- Data Modality
  - Modality
    - Tabular Data
- descriptiveStatistics
  - statistics
    - There were 220 document types used for the ground truth creation. 131 of them have a WW2-Thesaurus link, 85 of which contains a description.

3.2 Data Fields

- dataField
  - Data Field Name
    - Documenttype
  - Data Field Type
    - string
  - Description of the Data Field
    - This is the name of the document category.
- dataField
  - Data Field Name
    - URL thesaurus
  - Data Field Type
    - string
  - Description of the Data Field
    - URL from the WW2-Thesaurus representing this category.
  - usedVocabularies
    - Vocabulary
      - WW2-Thesaurus
    - Vocabulary Link
      - https://www.niod.nl/en/collections/ww2-thesaurus
- dataField
  - Data Field Name
    - Toelichting
  - Data Field Type
    - string
  - Description of the Data Field
    - Explanation of the category
- dataField
  - Data Field Name
    - Opmerking
  - Data Field Type
    - string
  - Description of the Data Field
    - Comment about the document type

3.3 Data Point Examples

- Typical Example
  - Description
    - Acte van invrijheidstelling

3.5 External Resources

- Resources
  - Yes

3.7 Social Impact, Sensitivity, and Biases

- Confidentiality
  - confidentialityBin
    - Yes
  - Specify Measures
    - We masked the scan filesnames, because they contain both the inventory number and the scan number. This can be traced to an individual.
- biases
  - Sensitive Attributes
    - Sensitive Human Attributes
      - None
- Ethical Review
  - Review
    - No

Level 3: Data

- 3.1 Data Resource Description
  - Name of Resource
    - Hierarchical structure document types GT
  - Description
    - This file shows how the document types are grouped in a hierarchy. This hierarchy was used during the ground truth creation, where the goal was to be as specific as possible.
  - Format
    - csv
  - Date
    - 2023-10-02
  - Language(s)
    - Dutch
  - Data Subject(s)
    - Subject(s)
      - Data about places and object
  - Data Modality
    - Modality
      - Tabular Data
- 3.2 Data Fields
  - dataField
    - Data Field Name
      - Volgorde
    - Data Field Type
      - integer
    - Description of the Data Field
      - Used to be able to sort the hierarchy in the main desired order.
  - dataField
    - Data Field Name
      - Subcategorie 1
    - Data Field Type
      - string
    - Description of the Data Field
      - First level of subcategories
  - dataField

Data Field Name
Subcategorie 2
Data Field Type
string
Description of the Data Field
Second level of subcategories
dataField
Data Field Name
Subcategorie 3
Data Field Type
string
Description of the Data Field
Third level of subcategories
dataField
Data Field Name
Subcategorie 4
Data Field Type
string
Description of the Data Field
Fourth and lowest level of subcategories
dataField
Data Field Name
Top-level
Data Field Type
string
Description of the Data Field
The top level categories. These can always be applied during ground truth creation, whereas it might not be possible to apply a subcategory.
3.5 External Resources
Resources
Yes
3.7 Social Impact, Sensitivity, and Biases
Safety Consideration
Does the resource contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?
No
Confidentiality
confidentialityBin
Yes
Specify Measures
We masked the scan filesnames, because they contain both the inventory number and the scan number. This can be traced to an individual.
Ethical Review
Review
No
Level 3: Data
3.1 Data Resource Description
Name of Resource

2023 Document classification GT anonymised - Statistics ground truth
Description
An overview of how many times the top-level and subcategories of document types were applied.
Format
csv
Size
1
Date
2023-10-02
Language(s)
Dutch
Data Subject(s)
Subject(s)
Data about places and object
Data Modality
Modality
Tabular Data
3.5 External Resources
Resources
Yes
3.7 Social Impact, Sensitivity, and Biases
Confidentiality
confidentialityBin
No
biases
Sensitive Attributes
Sensitive Human Attributes
None
Ethical Review
Review
No
Level 4: Uses
purposes
Production
Domain(s) of Application
Content Analysis
Domain(s) of Application
Machine Learning
Domain(s) of Application
Visual Analysis
Motivating Factor(s) and Problem Space(s)
This dataset is ground truth that was created in order to train a document classification model for the Central Archives of the Special Jurisdiction (CABR). This archive is held within the National Archives of the Netherlands. It contains files of around 425,000 people were investigated within the Special Jurisdiction. Among them were people who had been members of the National Socialist Movement (NSB), or who were suspected of crimes such as betrayal or fighting in the German army. But also people who were investigated and

found to be innocent. The archive contains around 30 million pages. Digitisation is ongoing and will finish in 2027. The archive contains a lot of different types of documents. They are also not grouped together within the files, and users are mostly interested in specific documents, such as verdicts and police reports. Furthermore, they have trouble reconstructing the judicial timeline, which document classification can facilitate. We therefore created this ground truth set to train a document classification model.

4.1. Uses

datasetUse

Safe for production use

Suitable Use Case(s)

Suitable Use Case

Getting an insight in what document types the files in the CABR consist of. ONLY RELEVANT ONCE SCANS BECOME AVAILABLE FOR TRAINING: Training a Machine Learning model to automatically classify document types in the CABR. Training a Machine Learning model to automatically classify the high-level document types found in this dataset (eg: handwritten documents, printed documents). Training a Machine Learning model to automatically classify either.

Additional Notes

The process of training the Machine Learning model based on this ground truth is ongoing.

Unsuitable Use Case(s)

unsuitableUseCase

Training outside of the dataset period and context

additionalNotes

The bulk of this ground truth are Dutch documents from the 1940’s and 1950’s. This is mostly evidence from before and during WW2, and legal documents from directly after WW2. Therefore, the ground truth is not ideal to train a set of documents from outside this period and context.

4.2. Use with Other Data

Safety Level

Conditionally safe to use with other data

Known safe dataset(s) or data type(s)

The ground truth provides some provisional insights in what the Central Archive of Special Jurisdiction consists of. If there is a comparable dataset of a related archive, it might be useful to look at the overlap and differences.

Known unsafe dataset(s) or Data Type(s)

It is not yet useful to use this dataset to train any kind of data, since the scans themselves are not available. The set of scans used for the ground truth (GT) is not representative for the entire CABR. At the time of the GT creation, the vast majority of scans were of the ‘Bijzondere Raad van Cassatie’, comparable to the

Supreme Court. There are collections of many other organisations in the CABR, which also contain different documents not available (enough) in the set we worked with.

4.3 Use in ML or AI Systems

Dataset Uses

Training

Notable Feature(s)

There are some relations between instances. They can belong to the same file, or to the same person in cases where there are multiple files about that person. This is not made explicit in the data, due to privacy issues.

Usage Guideline(s)

The bulk of this ground truth are Dutch documents from the 1940’s and 1950’s. This is mostly evidence from before and during WW2, and legal documents from directly after WW2. Therefore, the ground truth is not ideal to train a set of documents from outside this period and context. For now, it is also not possible to use this data to train a model based on scans of the Central Archive of Special Jurisdiction, since the scans themselves are not available due to privacy issues.

Level 5: Human Perspective

5.1. Human Annotators

Annotator description

Experts in World War II history

Annotator description

History generalists

Annotator description

Technical experts with practical knowledge of the archive

Annotator description

Archive experts with a focus on customer service

AnnotationType

Task Type

Annotate scans with categories

Description of annotators

Eight people were involved in creating the ground truth. Seven of them are salaried employees of the War in Court consortium partners (Huygens Institute: 3 | National Archives: 2 | WO2NET: 3). Three people were involved in checking a random sample of the created ground truth, and answering questions by the ground truth team. All of them are salaried employees of the National Archives that work with the CABR on a daily basis.

Compensation

All annotators are salaried employees, except for one person. They were a paid freelancer that worked for the Huygens Institute.

Language distribution of annotators

Dutch speakers

Age distribution of annotators

Twenties to early fifties

Gender distribution of annotators

5 out of 8 annotators identify as women, 3 as men.

Summary of annotation instructions

We instructed the annotators to: * Always provde a top-level category (eg: Empty page, Typed document). * If there were more top-level categories applicable, they could be added, but always in order of what's most applicable. For instance, if a document was primarily types with some hand-written notes added, the classification 'Typed document' was added first. * Then they needed to choose a specific document type, and as specific as possible. * Since it's easy to get stuck understanding what a document type is, the annotators were asked to not add a specific document type, but leave that empty. * Annotators could rite down questions or remarks in a specific column. This was then checked by the coordinator and solved, answered or forwarded to a colleague with the right expertise.

Annotation platform(s)

Google Spreadsheet; Internal website to view the scans

5.2. Creators and Contributors

Description of potential positionality impact factors

Eight people were involved in creating the ground truth. Seven of them are salaried employees of the War in Court consortium partners (Huygens Institute: 2 | National Archives: 2 | WO2NET: 3). One person was a paid freelancer and a World War II historian. Three people were involved in checking a random sample of the created ground truth, and answering questions by the ground truth team. All of them are salaried employees of the National Archives that work with the CABR on a daily basis.